SUPPLEMENTARY MATERIALS: AN EXPONENT RESTRICTION FOR ODD PERFECT NUMBERS NOT DIVISIBLE BY THREE*

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SM1. Guide to the computational verification. This supplement contains the eight Python programs, seven JSON records, README, SHA-256 manifest, and two legacy listing wrappers supplied in `OPN_verification.zip`. All nineteen files are reproduced below. The listings are embedded in this source; compilation requires `siamart251216.cls`, `OPN_shared.tex`, and the declared packages, but no external program or certificate files.

Python 3.8 or later is required to run the verification; no third-party Python packages are needed. Extract the archive and, from its `OPN_verification` directory, run

```
python3 verify_all.py
```

The driver checks seventeen manifest entries and runs five verification tasks in a temporary directory. The top-level `reproduction_receipt.json` records a fresh successful run of this package. The six other JSON records retain the archived computational results. The manifest excludes itself and the top-level reproduction receipt. File hashes establish byte consistency; the mathematical reductions and infinite-tail arguments remain in the article.

SM1.1. Guide to the listings. The README appears in Section SM2. Section SM3 contains the programs, beginning with the reproduction driver and proceeding through source verification, two-exception bounds, three-output enumeration, and seed exclusions. Section SM4 contains the corresponding JSON records. Section SM5 preserves the two original LaTeX wrappers as verbatim listings; they are not executed by this document. The main manuscript supplies the mathematical reductions and convergent tail arguments used by the finite calculations.

SM2. Package documentation. The README describes the submitted computational package and maps its programs to the article.

SM2.1. README.

README.md

```
# OPN computational verification package

Companion to "An Exponent Restriction for Odd Perfect Numbers Not Divisible by Three",
↪  M. R. DeDeo.

## Reproduction
Extract OPN_verification.zip, preserve its directory structure, and run from the
↪  OPN_verification directory:

    python3 verify_all.py

Python 3.8 or later is required. No third-party packages are needed. The tested
↪  interpreter is recorded in reproduction_receipt.json.
The driver checks 17 SHA-256 manifest entries, copies verification/ to a temporary
↪  directory, and runs five tasks there. Each task has a 120-second timeout. A
↪  timeout on slower hardware is not a mathematical counterexample. A successful run
↪  prints a JSON object with status PASS and five PASS check results. To retain a new
↪  receipt, redirect output to a new file, for example:

    python3 verify_all.py > new_reproduction_receipt.json

## Correspondence to the article
- verification/opn_pivot16/verify_residue_exclusion.py generates source factorizations
↪  and bounds (Lemma 2.4).
- verification/opn_audit17/check_certificate.py checks the source certificate; it is
↪  invoked by the two-exception verification task.
```

```
- verification/opn_stage18/verify_two_exception_bounds.py checks the finite bounds
↪  used in Proposition 3.1.
- verification/opn_stage19/verify_three_output_exclusion.py checks the finite source
↪  bounds, output enumeration, and tail comparisons (Lemmas 4.2-4.4).
- verification/opn_stage19/check_head_independently.py checks the finite maximum by a
↪  separate implementation (Lemma 4.3).
- verification/opn_review25/review_three_exception.py checks the enumeration using
↪  unordered output supports.
- verification/opn_stage21/verify_forced_families.py checks the seed inventory
↪  (Proposition 5.3).

Expected results include 907360 role-weighted configurations, maximum 538811/328320 at
↪  (q,r,s)=(13,7,31), and 99 excluded subsets with 17 inclusion-minimal members among
↪  511 nonempty subsets.

## Files and scope
This archive contains eight Python programs, seven JSON records (including
↪  reproduction_receipt.json), this README, MANIFEST.sha256, and two legacy listing
↪  wrappers: nineteen files in total.
The existing verification/ subdirectories are preserved because programs depend on
↪  relative paths. The six JSON records in those subdirectories are archived
↪  computation records. The top-level reproduction_receipt.json records the fresh run
↪  of this submission package.
MANIFEST.sha256 hashes seventeen files: it excludes itself and the top-level
↪  reproduction receipt. File hashes establish byte consistency, not mathematical
↪  correctness. The programs check finite calculations; mathematical reductions and
↪  infinite-tail arguments remain in the article. These checks are not external peer
↪  review.
OPN_py_listings.tex and OPN_json_listings.tex are optional legacy AMS listing
↪  wrappers; they are not used to run verification. The SIAM-formatted combined
↪  listings PDF and source are distributed separately.
```

SM3. Python programs.

SM3.1. Reproduction driver.

verify_all.py

```
"""Verify packaged hashes and reproduce exact arithmetic in a temporary copy.

This does not verify the analytic proofs or establish novelty.
"""

import hashlib,json,subprocess,sys,tempfile,shutil
from pathlib import Path
ROOT=Path(__file__).resolve().parent
manifest=ROOT/'MANIFEST.sha256'
if not manifest.exists():raise SystemExit('Missing MANIFEST.sha256')
checked=0
for line in manifest.read_text().splitlines():
    expected,name=line.split('  ',1)
    if hashlib.sha256((ROOT/name).read_bytes()).hexdigest()!=expected:
        raise SystemExit('Hash mismatch: '+name)
    checked+=1
jobs=[('two_exception','opn_stage18/verify_two_exception_bounds.py'),
      ('three_output','opn_stage19/verify_three_output_exclusion.py'),
      ('separate_head','opn_stage19/check_head_independently.py'),
      ('standalone_review','opn_review25/review_three_exception.py'),
      ('forced_families','opn_stage21/verify_forced_families.py')]
receipt={'status':'PASS','manifest_files':checked,'python':sys.version,'checks':[]}
with tempfile.TemporaryDirectory(prefix='tnp_reproduction_') as tmp:
    dst=Path(tmp)/'verification';shutil.copytree(ROOT/'verification',dst)
    for label,name in jobs:
        r=subprocess.run([sys.executable,str(dst/name)],text=True,capture_output=True
        ↪  ,timeout=120)
```

```
        if r.returncode:
            raise SystemExit(label+' failed:\n'+r.stdout+r.stderr)
        answer=json.loads(r.stdout)
        if answer.get('status')!='PASS':raise SystemExit(label+' did not pass')
        receipt['checks'].append({'name':label,'result':answer})
print(json.dumps(receipt,indent=2))
```

SM3.2. SHA-256 manifest.

```
MANIFEST.sha256
```

```
0ff06ff5cfa3e6a1427fe06c27d3ccafcab35056419eac44ca0c42805dd2c5fb  OPN_json_listings.tex
20703e72f37d031950ca13cfa16ac69301d35d6467f3216190e934c6801def3d  OPN_py_listings.tex
50e4479e9726cb93f17b6c5a0e8a5eac43375dcf1ff4a9dfc090054724249c99  README.md
242ddae49b5aa7d8fc317555905818f9e11d111094363254b8be55ac23e4034a
↪  verification/opn_audit17/check_certificate.py
014ca831ff044e56b092166e8a95fa353119df207e6c303a6207b3487441cfbd
↪  verification/opn_pivot16/residue_exclusion_certificate.json
fc0ccf9d3c733705f9a1a6b39dea239e6ed124216272f23aabff8b95970e7e44
↪  verification/opn_pivot16/verify_residue_exclusion.py
428a675d9574542b63a505f9f7578e7f4720405702b0ead2c03a2b818b0bb609
↪  verification/opn_review25/review_receipt.json
6776ca5437c8d46864f39c1d1fcab1254b79855efa223b15c5026b605a7ade41
↪  verification/opn_review25/review_three_exception.py
205872ca82b43e12972841840bb9f46eecd783807b45c4414c1dc203be13d33f
↪  verification/opn_stage18/two_exception_checks.json
e7c2ba13451217bc79ba17d406a8ab681b230351a59695a20879818aad9c0c8d
↪  verification/opn_stage18/verify_two_exception_bounds.py
a2e00d7d723c27e6810044cc87b59c21daa55d57abbd601a34d440cdb1d606c1
↪  verification/opn_stage19/check_head_independently.py
a5cbcb3ce887299a59c43519638f3a44023904b524a3f79a197523f4fcc02222
↪  verification/opn_stage19/independent_head_check.json
db961da551a4316eb410373e6c4efed572d0fc84d99b314abb8b060ad6c3dc4e
↪  verification/opn_stage19/three_output_certificate.json
a9cbc7ef8506c6b1f34217a5820b1368e81b8141c6b64516a51b77a34bd8b37b
↪  verification/opn_stage19/verify_three_output_exclusion.py
e6046bbe51b1a13a5a98e90ac17e3a5572989573736eedc634b6ae7bdf48bd28
↪  verification/opn_stage21/forced_family_certificate.json
ba4b2722ccb5731e584d1d205567162049e9509444c3e4a43e83b12d98cd8628
↪  verification/opn_stage21/verify_forced_families.py
18845cb45a6adad746993053e518392bee45b585cd54bf93fedbf8818d75b4aa  verify_all.py
```

SM3.3. Reconstructing the package. As an alternative to downloading the archive, the following command extracts all nineteen files from this LaTeX source into a new directory named `OPN_extracted`. Run it from the directory containing this source file. It preserves the original relative paths and refuses to overwrite an existing extraction directory.

```
python3 - <<'PYEXTRACT'
from pathlib import Path
import re
src = Path('OPN_supplement.tex')
text = src.read_text(encoding='utf-8')
pattern = (r'^% OPN-FILE-BEGIN ([^\n]+)\n'
           r'\\begin\{Verbatim\}\n(.*?)'
           r'^\\end\{Verbatim\}\n% OPN-FILE-END\n')
items = re.findall(pattern, text, re.M | re.S)
assert len(items) == 19, 'Expected nineteen embedded files'
dest = Path('OPN_extracted')
dest.mkdir(exist_ok=False)
for name, content in items:
    relative = Path(name)
    assert not relative.is_absolute() and '..' not in relative.parts
    target = dest / relative
```

```
    target.parent.mkdir(parents=True, exist_ok=True)
    target.write_bytes(content.encode('utf-8'))
print('Extracted nineteen files to', dest)
PYEXTRACT
cd OPN_extracted
python3 verify_all.py
```

SM3.4. Source enumeration and certificate generation.

verification/opn_pivot16/verify_residue_exclusion.py

```
"""Exact certificate for the finite part of the no-3 exponent-residue exclusion.
The infinite tail and prime-power lemma are proved/cited in the accompanying text.
Uses only Python standard library; no probabilistic primality or floating arithmetic.
"""
from fractions import Fraction
from math import isqrt, prod
from pathlib import Path
import json

LIMIT = 1000

def is_prime(n):
    return n >= 2 and all(n % t for t in range(2, isqrt(n) + 1))

def factor(n):
    ans = []
    t = 2
    while t*t <= n:
        e = 0
        while n % t == 0:
            n //= t
            e += 1
        if e:
            ans.append([t,e])
        t += 1
    if n > 1:
        ans.append([n,1])
    return ans

# Sieve independently specifies the complete list to be checked.
sieve = [True] * (LIMIT+1)
sieve[:2] = [False,False]
for t in range(2,isqrt(LIMIT)+1):
    if sieve[t]:
        for j in range(t*t,LIMIT+1,t):
            sieve[j] = False
primes = [p for p in range(5,LIMIT+1) if sieve[p] and p % 3 == 2]
rows = []
minima = {}
for p in primes:
    assert is_prime(p)
    value = p*p+p+1
    ff = factor(value)
    assert prod(r**e for r,e in ff) == value
    assert all(is_prime(r) and e > 0 for r,e in ff)
    assignments = []
    if len(ff) == 2:
        for j in range(2):
            q,a = ff[j]
            r,b = ff[1-j]
            if q % 12 == 1 and r % 3 == 1 and a+b <= 6:
                assignments.append({'q':q,'r':r,'a':a,'b':b})
                minima[(a,b)] = min(p,minima.get((a,b),p))
```

```
    rows.append({'p':p,'phi3':value,'factorization':ff,'assignments':assignments})
assert minima == {(1,1):29,(1,2):863}
C = Fraction(91,72)*Fraction(5,4)*Fraction(11,10)
table = []
for a in range(1,6):
    for b in range(1,7-a):
        lower = minima.get((a,b),LIMIT+1)
        table.append({'a':a,'b':b,'lower_bound':lower})
        C *= Fraction(lower,lower-1)
assert len(table) == 15
x = Fraction(2,5)
S = 2*x**7*(6/(1-x)+x/(1-x)**2)
assert S == Fraction(1024,28125)
assert C < Fraction(183,100)
assert C/(1-S) < Fraction(19,10) < 2

def rat(x):
    return {'numerator':x.numerator,'denominator':x.denominator}
result = {'limit':LIMIT,'checked_primes':len(primes),'rows':rows,'table':table,
          'finite_product':rat(C),'tail_log_bound':rat(S),
          'total_upper_bound':rat(C/(1-S)),
          'verified_total_less_than':'19/10',
          'scope':'Finite arithmetic certificate, not a bounded search for N.'}
out = Path(__file__).with_name('residue_exclusion_certificate.json')
out.write_text(json.dumps(result,indent=2)+'\n')
print(json.dumps({'checked_primes':len(primes),'low_patterns':len(minima),
                  'mixed_pairs':len(table),'total_bound':rat(C/(1-S)),
                  'all_exact_checks_passed':True}))
```

SM3.5. Source-certificate checker.

verification/opn_audit17/check_certificate.py

```
"""Read-only checker for Stage 16's finite certificate, Python standard library.

This implementation does not import or run the certificate generator. It checks
finite arithmetic, not the analytic proof or the published Diophantine input.
Usage: python3 check_certificate.py [path/to/residue_exclusion_certificate.json]
"""
import copy
import hashlib
import json
import sys
from fractions import Fraction as Q
from pathlib import Path


def require(condition, message):
    if not condition:
        raise ValueError(message)


def prime(n):
    if n < 2:
        return False
    if n in (2, 3):
        return True
    if n % 2 == 0 or n % 3 == 0:
        return False
    d = 5
    while d * d <= n:
        if n % d == 0 or n % (d + 2) == 0:
            return False
        d += 6
```

```
    return True


def rational(record):
    require(record['denominator'] > 0, 'nonpositive denominator')
    return Q(record['numerator'], record['denominator'])


def check(data):
    require(data['limit'] == 1000, 'wrong interval')
    expected = [p for p in range(5, 1001, 3) if prime(p)]
    require([row['p'] for row in data['rows']] == expected,
            'missing, duplicated, reordered, or nonprime input')
    require(data['checked_primes'] == len(expected) == 86, 'wrong count')
    minima = {}
    for row in data['rows']:
        p = row['p']
        value = p * p + p + 1
        require(row['phi3'] == value, 'wrong cyclotomic value')
        ff = row['factorization']
        bases = [pair[0] for pair in ff]
        require(bases == sorted(set(bases)), 'repeated or unordered bases')
        remaining = value
        for base, exponent in ff:
            require(prime(base) and type(exponent) is int and exponent > 0,
                    'invalid prime power')
            for _ in range(exponent):
                require(remaining % base == 0, 'factor does not divide')
                remaining //= base
        require(remaining == 1, 'incomplete factorization')
        assignments = []
        if len(ff) == 2:
            for (q, a), (r, b) in (ff, ff[::-1]):
                if q % 12 == 1 and r % 3 == 1 and a + b <= 6:
                    assignments.append(dict(q=q, r=r, a=a, b=b))
                    minima[(a, b)] = min(p, minima.get((a, b), p))
        require(row['assignments'] == assignments, 'wrong assignment list')
    require(minima == {(1, 1): 29, (1, 2): 863}, 'unexpected minima')
    bounds = {(a, j-a): minima.get((a, j-a), 1001)
              for j in range(2, 7) for a in range(1, j)}
    supplied = {(r['a'], r['b']): r['lower_bound'] for r in data['table']}
    require(len(data['table']) == 15 and supplied == bounds, 'wrong bound table')
    finite = Q(13, 12) * Q(7, 6) * Q(5, 4) * Q(11, 10)
    for lower in bounds.values():
        finite *= Q(lower, lower-1)
    # Sum all j>=2 first, then remove j=2,...,6: a different evaluation
    # from the shifted-tail expression in the generating program.
    x = Q(2, 5)
    tail = 2 * (x*x / (1-x)**2
                - sum((j-1) * x**j for j in range(2, 7)))
    total = finite / (1-tail)
    require(tail == Q(1024, 28125), 'tail arithmetic')
    require(rational(data['finite_product']) == finite, 'finite product mismatch')
    require(rational(data['tail_log_bound']) == tail, 'tail record mismatch')
    require(rational(data['total_upper_bound']) == total, 'total mismatch')
    require(finite < Q(183, 100), 'finite comparison failed')
    require(total < Q(183, 100) / (1-tail) < Q(19, 10) < 2,
            'final comparisons failed')
    require(data['verified_total_less_than'] == '19/10', 'bound label mismatch')
    return dict(inputs=len(expected), pairs=len(bounds),
                total_numerator=total.numerator, total_denominator=total.denominator)
```

```
def main():
    default = Path(__file__).resolve().parent.parent / 'opn_pivot16' /
    ↪  'residue_exclusion_certificate.json'
    path = Path(sys.argv[1]) if len(sys.argv) > 1 else default
    raw = path.read_bytes()
    data = json.loads(raw)
    result = check(data)
    # Negative controls guard against a checker that accepts damaged evidence.
    mutations = []
    for name in ('missing_input', 'wrong_factor', 'missing_assignment', 'wrong_tail'):
        bad = copy.deepcopy(data)
        if name == 'missing_input':
            bad['rows'].pop()
        elif name == 'wrong_factor':
            bad['rows'][0]['factorization'][0][0] += 2
        elif name == 'missing_assignment':
            next(row for row in bad['rows'] if row['assignments'])['assignments'] = []
        else:
            bad['tail_log_bound']['numerator'] += 1
        try:
            check(bad)
        except ValueError:
            mutations.append(name)
        else:
            raise ValueError('negative control accepted: ' + name)
    result.update(status='PASS', negative_controls_rejected=mutations,
                  certificate_sha256=hashlib.sha256(raw).hexdigest(),
                  checker_sha256=hashlib.sha256(Path(__file__).read_bytes()).hexdiges
                  ↪  t(),
                  scope='Finite arithmetic only; separate implementation, not external
                  ↪  review.')
    print(json.dumps(result, indent=2))


if __name__ == '__main__':
    main()
```

SM3.6. Two-exception bounds.

verification/opn_stage18/verify_two_exception_bounds.py

```
"""Exact arithmetic checks for the two-exception restriction.

Run from any directory. The project layout (also supplied in the ZIP) must
include opn_audit17/check_certificate.py and the Stage 16 JSON certificate.
This checks finite arithmetic, not the analytic proof or novelty.
"""
from fractions import Fraction as Q
from math import isqrt
from pathlib import Path
import hashlib
import json
import subprocess
import sys

root = Path(__file__).resolve().parent.parent
certificate = root / 'opn_pivot16' / 'residue_exclusion_certificate.json'
checker = root / 'opn_audit17' / 'check_certificate.py'
prior = json.loads(subprocess.check_output(
    [sys.executable, str(checker), str(certificate)], text=True))


def require(condition, message):
```

```
    if not condition:
        raise ValueError(message)


require(prior['status'] == 'PASS', 'prior certificate failed')
for p, factors in [(11, [7, 19]), (23, [7, 79]), (47, [37, 61])]:
    require(p*p+p+1 == factors[0]*factors[1], 'pure-source factorization')
    require(all(all(f % d for d in range(2, isqrt(f)+1)) for f in factors),
            'nonprime factor')
B = Q(19, 10) * Q(59, 58) / Q(11, 10)
require(B == Q(1121, 638) < Q(9, 5), 'two-output bound')
require(Q(13, 12)*Q(5, 4)*Q(11, 10)*Q(17, 16) < 2,
        'both exceptions 2 mod 3')
require(Q(9, 5)*Q(11, 10) == Q(99, 50) < 2, 'mixed case s>=11')
require(Q(9, 5)*Q(17, 20)*Q(5, 4) == Q(153, 80) < 2,
        'mixed case s=5')
require(Q(18, 5)**2 < 13 and Q(13, 5)**2 < 7, 'square root bounds')
require((1-Q(5, 18))*(1-Q(5, 13)) == Q(4, 9), 'geometric denominator')
tail = Q(9, 2*(40-1))
require(40**2 == 1600 and tail == Q(3, 26), 'third output tail')
upper = B * Q(1600, 1599) / (1-tail)
require(upper == Q(1793600, 902451) < 2, 'final exclusion bound')

# Candidate count by sieve, independently compared with trial division.
sieve = [True]*1600
sieve[0] = sieve[1] = False
for p in range(2, 40):
    if sieve[p]:
        for n in range(p*p, 1600, p):
            sieve[n] = False
candidates = [p for p in range(7, 1600, 3) if sieve[p]]
trial = [p for p in range(7, 1600) if p % 3 == 1
         and all(p % d for d in range(2, isqrt(p)+1))]
require(candidates == trial and len(candidates) == 121, 'candidate count')
pair_count = len(candidates)*(len(candidates)-1)//2
require(pair_count == 7260, 'pair count')
report = dict(status='PASS', prior_certificate_check=prior,
              two_output_bound=str(B), third_output_log_bound=str(tail),
              final_upper_bound=str(upper), exceptional_prime_candidates=candidates,
              candidate_count=len(candidates), unordered_pair_count=pair_count,
              checker_sha256=hashlib.sha256(Path(__file__).read_bytes()).hexdigest(),
              scope='Exact arithmetic checks; not formal proof verification or
              ↪  priority clearance.')
out = Path(__file__).with_name('two_exception_checks.json')
out.write_text(json.dumps(report, indent=2)+'\n')
print(json.dumps({k: report[k] for k in ('status', 'two_output_bound',
    'third_output_log_bound', 'final_upper_bound', 'candidate_count',
    ↪  'unordered_pair_count')}))
```

SM3.7. Three-output enumeration and tail bounds.

verification/opn_stage19/verify_three_output_exclusion.py

```
"""Exact finite certificate for the exclusion of two exceptional components.

Uses standard-library integer/rational arithmetic. The preceding proofs supply
the bounded output-prime ranges and the analytic infinite-tail argument.
Run: python3 opn_stage19/verify_three_output_exclusion.py
"""
from fractions import Fraction as F
from itertools import combinations
from math import isqrt
from pathlib import Path
import hashlib
```

```
import json
import subprocess
import sys

ROOT = Path(__file__).resolve().parent.parent
source = ROOT / 'opn_pivot16/residue_exclusion_certificate.json'
previous = json.loads(subprocess.check_output([sys.executable,
    str(ROOT/'opn_audit17/check_certificate.py'), str(source)], text=True))
rows = json.loads(source.read_text())['rows']


def require(test, message):
    if not test:
        raise ValueError(message)


def prime(n):
    return n >= 2 and all(n % d for d in range(2, isqrt(n)+1))


require(previous['status'] == 'PASS', 'input certificate failed')
# Two output primes both 7 modulo 12: complete reuse of certified factorizations.
minima = {}
for row in rows:
    ff = row['factorization']
    if len(ff) == 2 and all(v % 12 == 7 for v, e in ff):
        for (_, a), (_, b) in (ff, ff[::-1]):
            if a+b <= 6:
                minima[a, b] = min(row['p'], minima.get((a, b), row['p']))
require(minima == {(1, 1): 11, (1, 2): 521, (2, 1): 521}, 'mixed minima')
C7 = F(7, 6)*F(19, 18)*F(5, 4)*F(17, 16)
for a in range(1, 6):
    for b in range(1, 7-a):
        lower = minima.get((a, b), 1001)
        C7 *= F(lower, lower-1)
B7 = C7/(1-F(1024, 28125))
require(B7 < F(19, 10), '7-mod-12 kernel bound')
require(F(13, 5)**2 < 7 and F(13, 3)**2 < 19, 'root bounds')
require(F(8, 13)*F(10, 13) > F(4, 9), 'third output coefficient')
require(F(19, 10)*F(10000, 9999)*F(22, 21) == F(38000, 19089) < 2,
        'Euler cutoff bound')

# Complete output-prime lists, independently cross-checked by a sieve.
R = [p for p in range(7, 1600, 3) if prime(p)]
Q = [p for p in range(13, 10000, 12) if prime(p)]
sieve = [True]*10000
sieve[0] = sieve[1] = False
for p in range(2, 100):
    if sieve[p]:
        for n in range(p*p, 10000, p):
            sieve[n] = False
require(R == [p for p in range(7, 1600) if p % 3 == 1 and sieve[p]], 'R completeness')
require(Q == [p for p in range(13, 10000) if p % 12 == 1 and sieve[p]], 'Q
↪  completeness')
require(len(R) == 121 and len(Q) == 300, 'prime-list counts')

# Aggregate head factors by their exact prime support. No eligible row is lost.
support_factors = {}
for row in rows:
    support = tuple(v for v, e in row['factorization'])
    if len(support) <= 3:
        support_factors[support] = support_factors.get(support, F(1))*F(row['p'],
        ↪  row['p']-1)
```

```
base = {v: F(v, v-1)*support_factors.get((v,), F(1)) for v in set(R+Q)}
pair = {(u, v): support_factors.get(tuple(sorted((u, v))), F(1))
        for u in R for v in Q if u != v}
count = 0
maximum = F(0)
maximizer = None
for r, s in combinations(R, 2):
    fixed = base[r]*base[s]*support_factors.get((r, s), F(1))
    for q in Q:
        if q in (r, s):
            continue
        if q >= 1600 and (r % 12 == 1 or s % 12 == 1):
            continue
        head = fixed*base[q]*pair[r, q]*pair[s, q]
        head *= support_factors.get(tuple(sorted((q, r, s))), F(1))
        require(head < F(5, 3), 'head bound failed for ' + repr((q, r, s)))
        count += 1
        if head > maximum:
            maximum, maximizer = head, (q, r, s)
require(count == 907360, 'configuration count')
require(maximum == F(538811, 328320) and maximizer == (13, 7, 31), 'maximum')

# Independent formula for the number of configurations, not the nested-loop count.
require(count == sum(sum(q not in (r, s) for q in Q if
    q < 1600 or (r % 12 == 7 and s % 12 == 7)) for r, s in combinations(R, 2)),
    'configuration count cross-check')

# Rational upper bounds for reciprocal square roots of 7,13,19.
xs = [F(1000, isqrt(1000000*p)) for p in (7, 13, 19)]
for p, x in zip((7, 13, 19), xs):
    require(0 < x < 1 and p*x*x >= 1, 'reciprocal-root bound')
coeff = [F(1)]+[F(0)]*6
for x in xs:
    for j in range(1, 7):
        coeff[j] += x*coeff[j-1]
full = F(1)
for x in xs:
    full /= 1-x
delta = 2*(full-sum(coeff))
# Check the truncated coefficients against direct triple enumeration.
direct = sum(xs[0]**a*xs[1]**b*xs[2]**c
             for a in range(7) for b in range(7-a) for c in range(7-a-b))
require(sum(coeff) == direct and 0 < delta < F(3, 100), 'tail bound')
large_low_degree_slots = sum((j+1)*(j+2)//2 for j in range(1, 7))-3*5
require(large_low_degree_slots == 68, 'low-degree count')
K = F(1001, 1000)**68/(1-F(3, 100))
require(K < F(10, 9), 'large-source product bound')
require(F(5, 3)*F(10, 9) == F(50, 27) < 2, 'final contradiction')
report = dict(status='PASS', previous_certificate_check=previous,
    both_7_mod_12_minima=[dict(a=a, b=b, lower=p) for (a, b), p in
    ↪  sorted(minima.items())],
    both_7_mod_12_bound=str(B7), exceptional_candidates=R, euler_candidates=Q,
    checked_configurations=count, maximum_head=str(maximum),
    ↪  maximizing_q_r_s=maximizer,
    maximizing_small_sources=[row['p'] for row in rows
        if set(v for v,e in row['factorization']) <= set(maximizer)],
    reciprocal_root_upper_bounds=[str(x) for x in xs], tail_log_upper_bound=str(delta),
    low_degree_slots=68, large_source_product_bound=str(K), final_bound='50/27',
    checker_sha256=hashlib.sha256(Path(__file__).read_bytes()).hexdigest(),
    input_certificate_sha256=hashlib.sha256(source.read_bytes()).hexdigest(),
    scope='Exact finite arithmetic; analytic reductions require the accompanying
    ↪  proof.')
```

```
out = Path(__file__).with_name('three_output_certificate.json')
out.write_text(json.dumps(report, indent=2)+'\n')
print(json.dumps({k: report[k] for k in ('status', 'checked_configurations',
    'maximum_head', 'maximizing_q_r_s', 'maximizing_small_sources', 'final_bound')}))
```

SM3.8. Separate check of the finite maximum.

verification/opn_stage19/check_head_independently.py

```
"""Second exact head check: direct radical divisibility, no support-product cache.
Does not import or run the Stage 19 generator. Analytic claims are outside scope.
"""
from pathlib import Path
from itertools import combinations, chain
from math import prod
import hashlib
import json

root = Path(__file__).resolve().parent.parent
raw = (root/'opn_pivot16/residue_exclusion_certificate.json').read_bytes()
evidence = json.loads((root/'opn_stage19/three_output_certificate.json').read_text())
if hashlib.sha256(raw).hexdigest() != evidence['input_certificate_sha256']:
    raise ValueError('input certificate hash mismatch')
rows = json.loads(raw)['rows']
factors = [(prod(v for v,e in row['factorization']), row['p']) for row in rows]
sieve = bytearray(b'\x01')*10000
sieve[:2] = b'\x00\x00'
for p in range(2, 100):
    if sieve[p]:
        for m in range(p*p, 10000, p):
            sieve[m] = 0
R = [p for p in range(7,1600) if sieve[p] and p%3 == 1]
Q = [p for p in range(13,10000) if sieve[p] and p%12 == 1]
if R != evidence['exceptional_candidates'] or Q != evidence['euler_candidates']:
    raise ValueError('candidate list mismatch')
R7 = [r for r in R if r%12 == 7]
# Two disjoint case generators, instead of filtering the full triple loop.
small = ((q,r,s) for q in Q if q<1600 for r,s in combinations(R,2) if q!=r and q!=s)
large = ((q,r,s) for q in Q if q>=1600 for r,s in combinations(R7,2))
best_n, best_d, best_triple, count = 0, 1, None, 0
for q,r,s in chain(small,large):
    P = q*r*s
    num, den = P, (q-1)*(r-1)*(s-1)
    for radical,p in factors:
        if P % radical == 0:
            num *= p
            den *= p-1
    if 3*num >= 5*den:
        raise ValueError('head bound failed at '+repr((q,r,s)))
    if num*best_d > best_n*den:
        best_n, best_d, best_triple = num, den, (q,r,s)
    count += 1
if count != 907360 or 328320*best_n != 538811*best_d or best_triple != (13,7,31):
    raise ValueError('count or maximum mismatch')
print(json.dumps(dict(status='PASS', checked_configurations=count,
    maximum_head='538811/328320', maximizing_q_r_s=best_triple,
    method='Direct radical divisibility and integer cross multiplication',
    checker_sha256=hashlib.sha256(Path(__file__).read_bytes()).hexdigest(),
    scope='Separate implementation of finite head check; not external review.'),
    ↪  indent=2))
```

SM3.9. Enumeration by unordered output supports.

verification/opn_review25/review_three_exception.py

```
"""Fresh finite arithmetic review. No earlier project code/data are imported.
```

```
Analytic completeness and the published prime-power theorem are external proof
↪  obligations.
Python standard library only. Writes review_receipt.json alongside this script.
"""
from fractions import Fraction as F
from itertools import combinations
from math import isqrt, prod
from pathlib import Path
import hashlib,json,sys

def check(c,msg):
    if not c: raise ValueError(msg)
# Independently generate every prime needed for candidate lists and trial factors.
flags=bytearray(b'\1')*10000
flags[0]=flags[1]=0
for p in range(2,100):
    if flags[p]:
        flags[p*p:10000:p]=b'\0'*len(range(p*p,10000,p))
primes=[p for p in range(2,10000) if flags[p]]
def factor(n):
    out=[]
    for p in primes:
        if p*p>n: break
        a=0
        while n%p==0: n//=p;a+=1
        if a:out.append((p,a))
    if n>1:out.append((n,1))
    return out
rows=[]
for p in primes:
    if p<=1000 and p%3==2 and p!=2:
        f=factor(p*p+p+1)
        check(prod(v**a for v,a in f)==p*p+p+1,'recomposition')
        check(all(all(v%d for d in range(2,isqrt(v)+1)) for v,a in f),'prime factors')
        rows.append((p,f))
check(len(rows)==86,'source count')
minq={};min7={}
for p,f in rows:
    if len(f)==2:
        for (u,a),(v,b) in (f,list(reversed(f))):
            if a+b<=6:
                if u%12==1:minq[a,b]=min(p,minq.get((a,b),p))
                if u%12==v%12==7:min7[a,b]=min(p,min7.get((a,b),p))
check(minq=={(1,1):29,(1,2):863},'first minima')
check(min7=={(1,1):11,(1,2):521,(2,1):521},'second minima')
x=F(2,5)
delta=2*(x*x/(1-x)**2-sum((j-1)*x**j for j in range(2,7)))
check(delta==F(1024,28125),'two-output tail')
pairs=[(a,b) for a in range(1,6) for b in range(1,7-a)]
def pairproduct(table):
    return prod(F(table.get(ab,1001),table.get(ab,1001)-1) for ab in pairs)
C=F(13,12)*F(7,6)*F(5,4)*F(11,10)*pairproduct(minq)
check(C<F(183,100) and C/(1-delta)<F(19,10),'one exception')
B0=F(1121,638)
check(F(59,58)/F(11,10)*C/(1-delta)<B0<F(9,5),'two output kernel')
check(F(13,12)*F(5,4)*F(11,10)*F(17,16)<2,'two 2-mod3 exceptions')
check(F(9,5)*F(11,10)<2 and F(9,5)*F(17,20)*F(5,4)<2,'mixed exceptions')
check(F(18,5)**2<13 and F(13,5)**2<7,'root bounds 1')
check((1-F(5,18))*(1-F(5,13))==F(4,9),'root coefficient 1')
check(B0*F(1600,1599)*F(26,23)<2,'1600 bound')
B7=F(7,6)*F(19,18)*F(5,4)*F(17,16)*pairproduct(min7)/(1-delta)
check(B7<F(19,10),'two 7-mod12 kernel')
```

```
check(F(13,5)**2<7 and F(13,3)**2<19,'root bounds 2')
check(F(8,13)*F(10,13)>F(4,9),'root coefficient 2')
check(F(19,10)*F(10000,9999)*F(22,21)<2,'10000 bound')
R=[p for p in primes if p<1600 and p%3==1]
largeQ=[p for p in primes if p>=1600 and p%12==1]
R7=[p for p in R if p%12==7]
radicals=[(p,prod(v for v,a in f)) for p,f in rows]
best=(0,1);max_support=None;max_sources=None;unordered=0;roles=0

def examine(V,multiplicity):
    global best,max_support,max_sources,unordered,roles
    product=prod(V);num=product;den=prod(v-1 for v in V)
    source=[]
    for p,rad in radicals:
        if product%rad==0:num*=p;den*=p-1;source.append(p)
    check(3*num<5*den,'head upper bound')
    if num*best[1]>best[0]*den:best=(num,den);max_support=V;max_sources=source
    unordered+=1;roles+=multiplicity
# Below 1600 every 1-mod12 output may be designated Eulerian.
for V in combinations(R,3):
    k=sum(v%12==1 for v in V)
    if k:examine(V,k)
# A large output must be Eulerian; the two small outputs must be 7-mod12.
for q in largeQ:
    for r,s in combinations(R7,2):examine((r,s,q),1)
check(roles==907360,'role-weighted coverage')
check(F(*best)==F(538811,328320) and max_support==(7,13,31),'maximum')
check(max_sources==[5,191],'maximizing sources')
# Recompute the high-degree majorant explicitly, with no generating recurrence.
xs=[F(1000,isqrt(1000000*v)) for v in (7,13,19)]
check(all(0<x<1 and v*x*x>=1 for x,v in zip(xs,(7,13,19))),'reciprocal roots')
trunc=sum(xs[0]**a*xs[1]**b*xs[2]**c for a in range(7) for b in range(7-a) for c in
↪  range(7-a-b))
D=2*(prod(1/(1-x) for x in xs)-trunc)
low=[(a,b,c) for a in range(7) for b in range(7-a) for c in range(7-a-b) if a+b+c and
↪  not (sum(t>0 for t in (a,b,c))==1 and a+b+c>1)]
check(len(low)==68,'low degree slots')
check(0<D<F(3,100),'tail majorant')
K=F(1001,1000)**68/(1-F(3,100))
check(K<F(10,9) and F(5,3)*K<F(50,27)<2,'final bound')
result=dict(status='PASS',python=sys.version,source_count=len(rows),
    independent_unordered_supports=unordered,role_weighted_configurations=roles,
    maximum_head=str(F(*best)),maximizing_support=max_support,maximizing_sources=max_
    ↪  sources,
    two_output_tail=str(delta),two_output_kernel=str(B0),both_7_mod12_kernel=str(B7),
    tail_log_bound=str(D),low_degree_slots=len(low),large_source_bound=str(K),final_b
    ↪  ound='50/27',
    source_factorizations=[dict(p=p,factors=f) for p,f in rows],
    script_sha256=hashlib.sha256(Path(__file__).read_bytes()).hexdigest(),
    scope='Fresh finite arithmetic and rational comparisons; no prior certificate or
    ↪  script used. Not external or formal proof review.')
Path(__file__).with_name('review_receipt.json').write_text(json.dumps(result,indent=2
↪  )+'\n')
print(json.dumps({k:result[k] for k in ('status','independent_unordered_supports','ro
↪  le_weighted_configurations','maximum_head','maximizing_support','final_bound')}))
```

SM3.10. Forced seed families.

verification/opn_stage21/verify_forced_families.py

```
"""Exact one-step forced-support exclusions; standard library only.
Proves arithmetic certificates, not novelty. No unreviewed input is required.
"""
from fractions import Fraction as F
```

```
from itertools import combinations
from math import isqrt, prod
from pathlib import Path
import json

POOL=(5,11,17,23,29,41,47,53,59)


def prime(n):
    return n>=2 and all(n%d for d in range(2,isqrt(n)+1))


def factor(n):
    ans=[]
    d=2
    while d*d<=n:
        e=0
        while n%d==0:
            n//=d; e+=1
        if e: ans.append((d,e))
        d+=1
    if n>1: ans.append((n,1))
    return ans


def require(test, message):
    if not test: raise ValueError(message)


sources={}
for p in POOL:
    ff=factor(p*p+p+1)
    require(prime(p) and p%3==2, 'seed')
    require(prod(v**e for v,e in ff)==p*p+p+1 and all(prime(v) for v,e in ff),
    ↪  'factorization')
    sources[p]=ff
records=[]
by_seed={}
for size in range(1,len(POOL)+1):
    for S in combinations(POOL,size):
        P=sorted(set(S)|{v for p in S for v,e in sources[p]})
        base=prod(F(p*p+p+1,p*p) for p in P)
        candidates=[(None,base)]+[(q,base*F(q*(q+1),q*q+q+1)) for q in P if q%12==1]
        q,low=min(candidates,key=lambda z:z[1])
        # Recompute every Euler alternative directly, with unreduced integers.
        branches=[]
        for q2,value in candidates:
            facts=[(p,1 if p==q2 else 2) for p in P]
            D=prod(p**e for p,e in facts)
            sigma=prod(sum(p**j for j in range(e+1)) for p,e in facts)
            require(F(sigma,D)==value,'direct divisor arithmetic')
            branches.append({'euler_prime':q2,'divisor':D,'sigma_divisor':sigma,'inde
            ↪  x':str(value)})
        row={'seed':list(S),'forced_support':P,'distinct_forced_primes':len(P),
             'worst_euler_prime':q,'lower_bound':str(low),'excluded':low>2,'branches'
             ↪  :branches}
        records.append(row);by_seed[frozenset(S)]=row
minimal=[]
for row in records:
    S=frozenset(row['seed'])
    if row['excluded'] and not any(other < S and rr['excluded'] for other,rr in
    ↪  by_seed.items()):
```

```
        minimal.append(row)
require(len(records)==511 and len(minimal)==17,'inventory count')
for row in records:
    require(row['excluded']==any(set(m['seed'])<=set(row['seed']) for m in
    ↪  minimal),'minimal-family coverage')
require([m['seed'] for m in minimal if len(m['seed'])==4]==
    [[5,11,17,29],[5,11,23,29],[5,11,29,41],[5,11,29,47],[5,11,29,53]],'four-seed
    ↪  cases')
result={'status':'PASS','pool':POOL,'source_factorizations':sources,'checked_subsets'
↪  :len(records),
        'excluded_subsets':sum(r['excluded'] for r in
        ↪  records),'minimal_families':minimal,
        'all_subsets':records,'scope':'All subsets of the stated pool; minimal for
        ↪  this one-step lower-bound test only.'}
Path(__file__).with_name('forced_family_certificate.json').write_text(json.dumps(resu
↪  lt,indent=2)+'\n')
print(json.dumps({'status':'PASS','checked':len(records),'excluded':result['excluded_
↪  subsets'],
    'minimal':len(minimal),'forced_prime_count_range':[min(m['distinct_forced_primes']
    ↪  for m in minimal),max(m['distinct_forced_primes'] for m in minimal)]}))
```

SM4. JSON certificates and receipts.

SM4.1. Packaged reproduction receipt.

reproduction_receipt.json


```
{
  "status": "PASS",
  "manifest_files": 17,
  "python": "3.12.14 (main, Aug 25 2026, 14:00:49) [Clang 22.1.3 ]",
  "checks": [
    {
      "name": "two_exception",
      "result": {
        "status": "PASS",
        "two_output_bound": "1121/638",
        "third_output_log_bound": "3/26",
        "final_upper_bound": "1793600/902451",
        "candidate_count": 121,
        "unordered_pair_count": 7260
      }
    },
    {
      "name": "three_output",
      "result": {
        "status": "PASS",
        "checked_configurations": 907360,
        "maximum_head": "538811/328320",
        "maximizing_q_r_s": [
          13,
          7,
          31
        ],
        "maximizing_small_sources": [
          5,
          191
        ],
        "final_bound": "50/27"
      }
    },
    {
      "name": "separate_head",
      "result": {
        "status": "PASS",
        "checked_configurations": 907360,
        "maximum_head": "538811/328320",
        "maximizing_q_r_s": [
          13,
          7,
          31
        ],
        "method": "Direct radical divisibility and integer cross multiplication",
        "checker_sha256":
        ↪  "a2e00d7d723c27e6810044cc87b59c21daa55d57abbd601a34d440cdb1d606c1",
        "scope": "Separate implementation of finite head check; not external review."
      }
    },
    {
      "name": "standalone_review",
      "result": {
        "status": "PASS",
        "independent_unordered_supports": 751820,
        "role_weighted_configurations": 907360,
        "maximum_head": "538811/328320",
```

```
        "maximizing_support": [
          7,
          13,
          31
        ],
        "final_bound": "50/27"
      }
    },
    {
      "name": "forced_families",
      "result": {
        "status": "PASS",
        "checked": 511,
        "excluded": 99,
        "minimal": 17,
        "forced_prime_count_range": [
          10,
          17
        ]
      }
    }
  ]
}
```


SM4.2. Source factorizations and residue bounds.

verification/opn_pivot16/residue_exclusion_certificate.json


```
{
  "limit": 1000,
  "checked_primes": 86,
  "rows": [
    {
      "p": 5,
      "phi3": 31,
      "factorization": [
        [
          31,
          1
        ]
      ],
      "assignments": []
    },
    {
      "p": 11,
      "phi3": 133,
      "factorization": [
        [
          7,
          1
        ],
        [
          19,
          1
        ]
      ],
      "assignments": []
    },
    {
      "p": 17,
      "phi3": 307,
      "factorization": [
        [
          307,
```

```
      1
    ]
  ],
  "assignments": []
},
{
  "p": 23,
  "phi3": 553,
  "factorization": [
    [
      7,
      1
    ],
    [
      79,
      1
    ]
  ],
  "assignments": []
},
{
  "p": 29,
  "phi3": 871,
  "factorization": [
    [
      13,
      1
    ],
    [
      67,
      1
    ]
  ],
  "assignments": [
    {
      "q": 13,
      "r": 67,
      "a": 1,
      "b": 1
    }
  ]
},
{
  "p": 41,
  "phi3": 1723,
  "factorization": [
    [
      1723,
      1
    ]
  ],
  "assignments": []
},
{
  "p": 47,
  "phi3": 2257,
  "factorization": [
    [
      37,
      1
    ],
    [
```

```
        61,
        1
      ]
    ],
    "assignments": [
      {
        "q": 37,
        "r": 61,
        "a": 1,
        "b": 1
      },
      {
        "q": 61,
        "r": 37,
        "a": 1,
        "b": 1
      }
    ]
  },
  {
    "p": 53,
    "phi3": 2863,
    "factorization": [
      [
        7,
        1
      ],
      [
        409,
        1
      ]
    ],
    "assignments": [
      {
        "q": 409,
        "r": 7,
        "a": 1,
        "b": 1
      }
    ]
  },
  {
    "p": 59,
    "phi3": 3541,
    "factorization": [
      [
        3541,
        1
      ]
    ],
    "assignments": []
  },
  {
    "p": 71,
    "phi3": 5113,
    "factorization": [
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      "lower_bound": 1001
    },
    {
      "a": 4,
      "b": 1,
      "lower_bound": 1001
    },
    {
      "a": 4,
      "b": 2,
      "lower_bound": 1001
    },
    {
      "a": 5,
      "b": 1,
      "lower_bound": 1001
    }
  ],
  "finite_product": {
    "numerator": 362566637027574375657841003001644844368677186 1,
    "denominator": 198604800000000000000000000000000000000000000000000
  },
  "tail_log_bound": {
    "numerator": 1024,
    "denominator": 28125
  },
  "total_upper_bound": {
    "numerator": 362566637027574375657841003001644844368677186 1,
    "denominator": 191373819904000000000000000000000000000000000000000
  },
  "verified_total_less_than": "19/10",
  "scope": "Finite arithmetic certificate, not a bounded search for N."
}
```


SM4.3. Two-exception comparisons.

verification/opn_stage18/two_exception_checks.json


```
{
  "status": "PASS",
  "prior_certificate_check": {
    "inputs": 86,
    "pairs": 15,
    "total_numerator": 362566637027574375657841003001644844368677186 1,
    "total_denominator": 191373819904000000000000000000000000000000000000000,
    "status": "PASS",
    "negative_controls_rejected": [
      "missing_input",
      "wrong_factor",
      "missing_assignment",
      "wrong_tail"
    ],
    "certificate_sha256":
    ↪  "014ca831ff044e56b092166e8a95fa353119df207e6c303a6207b3487441cfbd",
    "checker_sha256":
    ↪  "242ddae49b5aa7d8fc317555905818f9e11d111094363254b8be55ac23e4034a",
    "scope": "Finite arithmetic only; separate implementation, not external review."
  },
  "two_output_bound": "1121/638",
  "third_output_log_bound": "3/26",
  "final_upper_bound": "1793600/902451",
```

```
"exceptional_prime_candidates": [
  7,
  13,
  19,
  31,
  37,
  43,
  61,
  67,
  73,
  79,
  97,
  103,
  109,
  127,
  139,
  151,
  157,
  163,
  181,
  193,
  199,
  211,
  223,
  229,
  241,
  271,
  277,
  283,
  307,
  313,
  331,
  337,
  349,
  367,
  373,
  379,
  397,
  409,
  421,
  433,
  439,
  457,
  463,
  487,
  499,
  523,
  541,
  547,
  571,
  577,
  601,
  607,
  613,
  619,
  631,
  643,
  661,
  673,
  691,
  709,
  727,
```

```
    733,
    739,
    751,
    757,
    769,
    787,
    811,
    823,
    829,
    853,
    859,
    877,
    883,
    907,
    919,
    937,
    967,
    991,
    997,
    1009,
    1021,
    1033,
    1039,
    1051,
    1063,
    1069,
    1087,
    1093,
    1117,
    1123,
    1129,
    1153,
    1171,
    1201,
    1213,
    1231,
    1237,
    1249,
    1279,
    1291,
    1297,
    1303,
    1321,
    1327,
    1381,
    1399,
    1423,
    1429,
    1447,
    1453,
    1459,
    1471,
    1483,
    1489,
    1531,
    1543,
    1549,
    1567,
    1579,
    1597
  ],
  "candidate_count": 121,
```

```
  "unordered_pair_count": 7260,
  "checker_sha256": "e7c2ba13451217bc79ba17d406a8ab681b230351a59695a20879818aad9c0c8d",
  "scope": "Exact arithmetic checks; not formal proof verification or priority
  ↪  clearance."
}
```


SM4.4. Three-output certificate.

verification/opn_stage19/three_output_certificate.json


```
{
  "status": "PASS",
  "previous_certificate_check": {
    "inputs": 86,
    "pairs": 15,
    "total_numerator": 362566637027574375657841003001644844368677186 1,
    "total_denominator": 191373819904000000000000000000000000000000000000,
    "status": "PASS",
    "negative_controls_rejected": [
      "missing_input",
      "wrong_factor",
      "missing_assignment",
      "wrong_tail"
    ],
    "certificate_sha256":
    ↪  "014ca831ff044e56b092166e8a95fa353119df207e6c303a6207b3487441cfbd",
    "checker_sha256":
    ↪  "242ddae49b5aa7d8fc317555905818f9e11d111094363254b8be55ac23e4034a",
    "scope": "Finite arithmetic only; separate implementation, not external review."
  },
  "both_7_mod_12_minima": [
    {
      "a": 1,
      "b": 1,
      "lower": 11
    },
    {
      "a": 1,
      "b": 2,
      "lower": 521
    },
    {
      "a": 2,
      "b": 1,
      "lower": 521
    }
  ],
  "both_7_mod_12_bound": "40428806363919721730434751568577497279209119/21313093632000
  ↪  0000000000000000000000000000000",
  "exceptional_candidates": [
    7,
    13,
    19,
    31,
    37,
    43,
    61,
    67,
    73,
    79,
    97,
    103,
    109,
    127,
```

```
139,
151,
157,
163,
181,
193,
199,
211,
223,
229,
241,
271,
277,
283,
307,
313,
331,
337,
349,
367,
373,
379,
397,
409,
421,
433,
439,
457,
463,
487,
499,
523,
541,
547,
571,
577,
601,
607,
613,
619,
631,
643,
661,
673,
691,
709,
727,
733,
739,
751,
757,
769,
787,
811,
823,
829,
853,
859,
877,
883,
907,
919,
```

```
    937,
    967,
    991,
    997,
    1009,
    1021,
    1033,
    1039,
    1051,
    1063,
    1069,
    1087,
    1093,
    1117,
    1123,
    1129,
    1153,
    1171,
    1201,
    1213,
    1231,
    1237,
    1249,
    1279,
    1291,
    1297,
    1303,
    1321,
    1327,
    1381,
    1399,
    1423,
    1429,
    1447,
    1453,
    1459,
    1471,
    1483,
    1489,
    1531,
    1543,
    1549,
    1567,
    1579,
    1597
  ],
  "euler_candidates": [
    13,
    37,
    61,
    73,
    97,
    109,
    157,
    181,
    193,
    229,
    241,
    277,
    313,
    337,
    349,
```

```
373,
397,
409,
421,
433,
457,
541,
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613,
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769,
829,
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877,
937,
997,
1009,
1021,
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1069,
1093,
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1129,
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1201,
1213,
1237,
1249,
1297,
1321,
1381,
1429,
1453,
1489,
1549,
1597,
1609,
1621,
1657,
1669,
1693,
1741,
1753,
1777,
1789,
1801,
1861,
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1993,
2017,
2029,
2053,
2089,
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2137,
2161,
```

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2677,
2689,
2713,
2749,
2797,
2833,
2857,
2917,
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3049,
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3109,
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3169,
3181,
3217,
3229,
3253,
3301,
3313,
3361,
3373,
3433,
3457,
3469,
3517,
3529,
3541,
3613,
3637,
3673,
3697,
3709,
3733,
3769,
3793,
3853,
3877,
3889,
4021,
4057,
4093,
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4153,
4177,
4201,
4261,
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```
4273,
4297,
4357,
4441,
4513,
4549,
4561,
4597,
4621,
4657,
4729,
4789,
4801,
4813,
4861,
4909,
4933,
4957,
4969,
4993,
5077,
5101,
5113,
5197,
5209,
5233,
5281,
5413,
5437,
5449,
5521,
5557,
5569,
5581,
5641,
5653,
5689,
5701,
5737,
5749,
5821,
5857,
5869,
5881,
5953,
6037,
6073,
6121,
6133,
6217,
6229,
6277,
6301,
6337,
6361,
6373,
6397,
6421,
6469,
6481,
6529,
6553,
```

```
6577,
6637,
6661,
6673,
6709,
6733,
6781,
6793,
6829,
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6949,
6961,
6997,
7057,
7069,
7129,
7177,
7213,
7237,
7297,
7309,
7321,
7333,
7369,
7393,
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7477,
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7537,
7549,
7561,
7573,
7621,
7669,
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7717,
7741,
7753,
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7993,
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8089,
8101,
8161,
8209,
8221,
8233,
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8317,
8329,
8353,
8377,
8389,
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8521,
8581,
8629,
8641,
```

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    8677,
    8689,
    8713,
    8737,
    8761,
    8821,
    8893,
    8929,
    8941,
    9001,
    9013,
    9049,
    9109,
    9133,
    9157,
    9181,
    9241,
    9277,
    9337,
    9349,
    9397,
    9421,
    9433,
    9601,
    9613,
    9649,
    9661,
    9697,
    9721,
    9733,
    9769,
    9781,
    9817,
    9829,
    9901,
    9949,
    9973
  ],
  "checked_configurations": 907360,
  "maximum_head": "538811/328320",
  "maximizing_q_r_s": [
    13,
    7,
    31
  ],
  "maximizing_small_sources": [
    5,
    191
  ],
  "reciprocal_root_upper_bounds": [
    "200/529",
    "200/721",
    "500/2179"
  ],
  "tail_log_upper_bound": "17074392539982939992703464726824171509509201000000000000000
  ↪  0/589043602236810470739900741214086727290275449810580802098311",
  "low_degree_slots": 68,
```

```
  "large_source_product_bound": "10703289409195582294861258757518518204200330543322496592220226699092798685664573063029667643056843840059785047479799971450431738533800458436331528333145561291534781394914956660993213677689424351182780680019700000000000
```

```
  "final_bound": "50/27",
  "source_factorizations": [
    {
      "p": 5,
      "factors": [
        [
          31,
          1
        ]
      ]
    },
    {
      "p": 11,
      "factors": [
        [
          7,
          1
        ],
        [
          19,
          1
        ]
      ]
    },
    {
      "p": 17,
      "factors": [
        [
          307,
          1
        ]
      ]
    },
    {
      "p": 23,
      "factors": [
        [
          7,
          1
        ],
        [
          79,
          1
        ]
      ]
    },
    {
      "p": 29,
      "factors": [
        [
          13,
          1
        ],
        [
          67,
          1
        ]
      ]
    },
    {
      "p": 41,
      "factors": [
```

```
    [
      1723,
      1
    ]
  ]
},
{
  "p": 47,
  "factors": [
    [
      37,
      1
    ],
    [
      61,
      1
    ]
  ]
},
{
  "p": 53,
  "factors": [
    [
      7,
      1
    ],
    [
      409,
      1
    ]
  ]
},
{
  "p": 59,
  "factors": [
    [
      3541,
      1
    ]
  ]
},
{
  "p": 71,
  "factors": [
    [
      5113,
      1
    ]
  ]
},
{
  "p": 83,
  "factors": [
    [
      19,
      1
    ],
    [
      367,
      1
    ]
  ]
```

```
    },
    {
      "p": 89,
      "factors": [
        [
          8011,
          1
        ]
      ]
    },
    {
      "p": 101,
      "factors": [
        [
          10303,
          1
        ]
      ]
    },
    {
      "p": 107,
      "factors": [
        [
          7,
          1
        ],
        [
          13,
          1
        ],
        [
          127,
          1
        ]
      ]
    },
    {
      "p": 113,
      "factors": [
        [
          13,
          1
        ],
        [
          991,
          1
        ]
      ]
    },
    {
      "p": 131,
      "factors": [
        [
          17293,
          1
        ]
      ]
    },
    {
      "p": 137,
      "factors": [
        [
```

```
      7,
      1
    ],
    [
      37,
      1
    ],
    [
      73,
      1
    ]
  ]
},
{
  "p": 149,
  "factors": [
    [
      7,
      1
    ],
    [
      31,
      1
    ],
    [
      103,
      1
    ]
  ]
},
{
  "p": 167,
  "factors": [
    [
      28057,
      1
    ]
  ]
},
{
  "p": 173,
  "factors": [
    [
      30103,
      1
    ]
  ]
},
{
  "p": 179,
  "factors": [
    [
      7,
      1
    ],
    [
      4603,
      1
    ]
  ]
},
{
```

```
      "p": 191,
      "factors": [
        [
          7,
          1
        ],
        [
          13,
          2
        ],
        [
          31,
          1
        ]
      ]
    },
    {
      "p": 197,
      "factors": [
        [
          19,
          1
        ],
        [
          2053,
          1
        ]
      ]
    },
    {
      "p": 227,
      "factors": [
        [
          73,
          1
        ],
        [
          709,
          1
        ]
      ]
    },
    {
      "p": 233,
      "factors": [
        [
          7,
          1
        ],
        [
          7789,
          1
        ]
      ]
    },
    {
      "p": 239,
      "factors": [
        [
          19,
          1
        ],
```

```
      [
        3019,
        1
      ]
    ]
  },
  {
    "p": 251,
    "factors": [
      [
        43,
        1
      ],
      [
        1471,
        1
      ]
    ]
  },
  {
    "p": 257,
    "factors": [
      [
        61,
        1
      ],
      [
        1087,
        1
      ]
    ]
  },
  {
    "p": 263,
    "factors": [
      [
        7,
        2
      ],
      [
        13,
        1
      ],
      [
        109,
        1
      ]
    ]
  },
  {
    "p": 269,
    "factors": [
      [
        13,
        1
      ],
      [
        37,
        1
      ],
      [
        151,
```

```
        1
      ]
    ]
  },
  {
    "p": 281,
    "factors": [
      [
        109,
        1
      ],
      [
        727,
        1
      ]
    ]
  },
  {
    "p": 293,
    "factors": [
      [
        86143,
        1
      ]
    ]
  },
  {
    "p": 311,
    "factors": [
      [
        19,
        1
      ],
      [
        5107,
        1
      ]
    ]
  },
  {
    "p": 317,
    "factors": [
      [
        7,
        1
      ],
      [
        14401,
        1
      ]
    ]
  },
  {
    "p": 347,
    "factors": [
      [
        7,
        1
      ],
      [
        13,
        1
```

```
    ],
    [
      1327,
      1
    ]
  ]
},
{
  "p": 353,
  "factors": [
    [
      19,
      1
    ],
    [
      6577,
      1
    ]
  ]
},
{
  "p": 359,
  "factors": [
    [
      7,
      1
    ],
    [
      37,
      1
    ],
    [
      499,
      1
    ]
  ]
},
{
  "p": 383,
  "factors": [
    [
      147073,
      1
    ]
  ]
},
{
  "p": 389,
  "factors": [
    [
      7,
      1
    ],
    [
      21673,
      1
    ]
  ]
},
{
  "p": 401,
  "factors": [
```

```
      [
        7,
        1
      ],
      [
        23029,
        1
      ]
    ]
  },
  {
    "p": 419,
    "factors": [
      [
        13,
        1
      ],
      [
        13537,
        1
      ]
    ]
  },
  {
    "p": 431,
    "factors": [
      [
        7,
        1
      ],
      [
        67,
        1
      ],
      [
        397,
        1
      ]
    ]
  },
  {
    "p": 443,
    "factors": [
      [
        7,
        1
      ],
      [
        28099,
        1
      ]
    ]
  },
  {
    "p": 449,
    "factors": [
      [
        97,
        1
      ],
      [
        2083,
```

```
      1
    ]
  ]
},
{
  "p": 461,
  "factors": [
    [
      373,
      1
    ],
    [
      571,
      1
    ]
  ]
},
{
  "p": 467,
  "factors": [
    [
      19,
      1
    ],
    [
      11503,
      1
    ]
  ]
},
{
  "p": 479,
  "factors": [
    [
      43,
      1
    ],
    [
      5347,
      1
    ]
  ]
},
{
  "p": 491,
  "factors": [
    [
      37,
      1
    ],
    [
      6529,
      1
    ]
  ]
},
{
  "p": 503,
  "factors": [
    [
      13,
      1
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```
      ],
      [
        19501,
        1
      ]
    ]
  },
  {
    "p": 509,
    "factors": [
      [
        43,
        1
      ],
      [
        6037,
        1
      ]
    ]
  },
  {
    "p": 521,
    "factors": [
      [
        31,
        2
      ],
      [
        283,
        1
      ]
    ]
  },
  {
    "p": 557,
    "factors": [
      [
        7,
        2
      ],
      [
        6343,
        1
      ]
    ]
  },
  {
    "p": 563,
    "factors": [
      [
        31,
        1
      ],
      [
        10243,
        1
      ]
    ]
  },
  {
    "p": 569,
    "factors": [
```

```
      [
        7,
        2
      ],
      [
        6619,
        1
      ]
    ]
  },
  {
    "p": 587,
    "factors": [
      [
        547,
        1
      ],
      [
        631,
        1
      ]
    ]
  },
  {
    "p": 593,
    "factors": [
      [
        163,
        1
      ],
      [
        2161,
        1
      ]
    ]
  },
  {
    "p": 599,
    "factors": [
      [
        7,
        1
      ],
      [
        51343,
        1
      ]
    ]
  },
  {
    "p": 617,
    "factors": [
      [
        97,
        1
      ],
      [
        3931,
        1
      ]
    ]
  },
```

```
{
  "p": 641,
  "factors": [
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      7,
      1
    ],
    [
      58789,
      1
    ]
  ]
},
{
  "p": 647,
  "factors": [
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      211,
      1
    ],
    [
      1987,
      1
    ]
  ]
},
{
  "p": 653,
  "factors": [
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SM4.7. Complete forced-family certificate.

verification/opn_stage21/forced_family_certificate.json


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    "excluded": true,
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      "divisor":
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      "sigma_divisor":
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      "index":
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    }
  ]
},
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  "seed": [
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    17,
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    59
```

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      "sigma_divisor":
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      "sigma_divisor":
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      "index":
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      "index":
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    }
  ]
},
{
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  "excluded": true,
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          ↪  2267023304354797,
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          ↪  537914351152826,
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          ↪  3836435103941273775"
        },
        {
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          "divisor":
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          ↪  2410510687935497425"
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          ↪  337695"
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          "divisor":
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          "sigma_divisor":
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          ↪  111975"
        }
      ]
    }
  ],
  "scope": "All subsets of the stated pool; minimal for this one-step lower-bound test
  ↪  only."
}
```

SM5. Legacy listing wrappers. These legacy wrappers use external file inputs. They are retained for completeness and are not required for verification or for compiling this SIAM supplement.

SM5.1. Original Python wrapper.

OPN_py_listings.tex

```
\documentclass[11pt]{amsart}
\usepackage[T1]{fontenc}
\usepackage{fvextra}
\usepackage[margin=1in]{geometry}
\begin{document}
\title{Odd perfect numbers: Python programs}
\maketitle
The listings below reproduce the accompanying machine-readable files.
\section*{\texttt{\detokenize{verification/opn_audit17/check_certificate.py}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_audit17/check_certificate.py}
\section*{\texttt{\detokenize{verification/opn_pivot16/verify_residue_exclusion.py}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_pivot16/verify_residue_exclusion.py}
\section*{\texttt{\detokenize{verification/opn_review25/review_three_exception.py}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_review25/review_three_exception.py}
\section*{\texttt{\detokenize{verification/opn_stage18/verify_two_exception_bounds.py
↪  }}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_stage18/verify_two_exception_bounds.py}
\section*{\texttt{\detokenize{verification/opn_stage19/check_head_independently.py}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_stage19/check_head_independently.py}
\section*{\texttt{\detokenize{verification/opn_stage19/verify_three_output_exclusion.
↪  py}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_stage19/verify_three_output_exclusion.py}
\section*{\texttt{\detokenize{verification/opn_stage21/verify_forced_families.py}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_stage21/verify_forced_families.py}
\section*{\texttt{\detokenize{verify_all.py}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verify_all.py}
\end{document}
```

SM5.2. Original JSON wrapper.

OPN_json_listings.tex

```
\documentclass[11pt]{amsart}
\usepackage[T1]{fontenc}
\usepackage{fvextra}
\usepackage[margin=1in]{geometry}
\begin{document}
\title{Odd perfect numbers: JSON certificates}
\maketitle
The listings below reproduce the accompanying machine-readable files.
\section*{\texttt{\detokenize{verification/opn_pivot16/residue_exclusion_certificate.
↪  json}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_pivot16/residue_exclusion_certificate.json}
\section*{\texttt{\detokenize{verification/opn_review25/review_receipt.json}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_review25/review_receipt.json}
\section*{\texttt{\detokenize{verification/opn_stage18/two_exception_checks.json}}}
```

```
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_stage18/two_exception_checks.json}
\section*{\texttt{\detokenize{verification/opn_stage19/independent_head_check.json}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_stage19/independent_head_check.json}
\section*{\texttt{\detokenize{verification/opn_stage19/three_output_certificate.json}}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_stage19/three_output_certificate.json}
\section*{\texttt{\detokenize{verification/opn_stage21/forced_family_certificate.json
↪  }}}
\VerbatimInput[fontsize=\scriptsize,breaklines=true,breakanywhere=true]{verification/
↪  opn_stage21/forced_family_certificate.json}
\end{document}
```